Unsupervised Occluded Target Detection Based on Spherical Shell With Multispectral Point Clouds

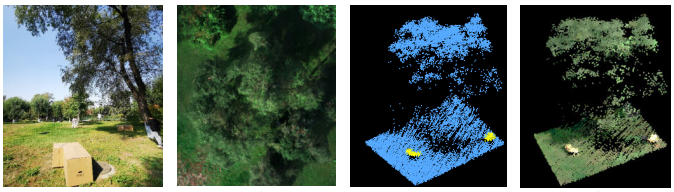
Likun Chen, *Member, IEEE*, Yanfeng Gu, *Senior Member, IEEE*, and Xian Li, *Member, IEEE*

***Abstract*— Multispectral point clouds (MPCs) acquired from uncrewed aerial vehicles (UAVs), leveraging LiDAR's canopy-penetrating capacity, provide distinct advantages for detecting occluded targets beneath vegetation canopy. However, limited samples, missing target spatial morphology, and unstructured data format have led to low occluded detection accuracy. Given these constraints, an unsupervised stereo detection method for occluded targets based on the spherical shell model with MPCs from UAVs has been proposed for the first time. The method exploits the spatial–spectral differences between targets and the background, treating targets as anomalies within the background, and enables occluded target detection without requiring training. The spherical shell model has been constructed in MPCs to avoid contaminating the global background with targets, leveraging its local separation characteristics for unsupervised detection of occluded targets. An adaptive radius and multiscale spatial feature extraction have been designed to enhance the method's robustness. The collaborative representation model has been utilized to achieve unsupervised detection by suppressing expressible points in local background features. To bridge the gap between algorithmic metrics and practical requirements, we have further proposed a target-level evaluation method that overcomes the limitations of conventional methods, which are susceptible to density-induced false alarm distortions in unstructured MPCs. Experiments on three real-world MPCs and a public airborne hyperspectral and LiDAR dataset show that our method achieves higher detection accuracy than existing spectral-based detectors and performs better on exposed targets.**

***Index Terms*— Multispectral point cloud (MPC), occluded target, uncrewed aerial vehicles (UAVs), unsupervised target detection.**

I. Introduction

THE multispectral point clouds (MPCs) obtained from uncrewed aerial vehicle (UAV) platforms record both spatial information and spectral information of the land covers, allowing target detection to move beyond reliance on spatial information solely from single-band point clouds for tasks such as instance segmentation [1], [2] and semantic segmentation [3], [4], making MPC detection gradually become a growing focus of research. However, in natural scenes, target occlusion by vegetation is common, resulting in low exposure probability and missing spatial morphology for occluded targets in orthophotos, making detection challenging. Compared to multispectral and hyperspectral orthophotos obtained by UAVs, MPCs, with the strong directional capability and active imaging of the LiDAR sensor, can penetrate canopy gaps and capture information of partially occluded targets [5] [Fig. 1(b)] and have the potential for occluded targets detection. Compared to the SAR sensors, LiDAR sensors on UAVs cannot penetrate leaves. However, by exploiting the gaps between leaves and leveraging a large scanning angle, LiDAR can capture spatial information of objects occluded by the canopy. A higher spatial resolution, stronger resistance to active interference, faster data acquisition, and no blind zones caused by complex object echoes give LiDAR an advantage in detecting occluded targets under the forest canopy.



(a) (b) (c) (d)

Fig. 1. Compared to orthophoto imagery, MPC data can capture more information about land covers that are occluded by the canopy. (a) Targets occluded by the canopy. (b) Targets are occluded in the orthophoto. (c) Ground truth (side view). (d) MPC data from UAV can record the occluded land covers (side view).

Occluded targets in current UAV laser scanning (ULS) applications have received limited attention due to a lack of target prior information and missing spatial morphology, as mentioned above. Existing studies have focused on detecting vehicles [6], coarse woody debris [7], and tree trunks beneath the canopy [8], as well as capturing high-resolution topography [9] and ancient microtopography [10] under occlusion conditions. The above methods demonstrate the ability of single-band ULS point cloud data to detect occluded targets but rely on extensive sample training. However, in the task of stereo target detection under vegetation canopy occlusion in complex land covers using MPCs from UAV, we face the following challenge: 1) canopy occlusion leads to a lack of target priors, low exposure, incomplete morphology, and spectral variability [11], [12], making sample-based training challenging and 2) research on unsupervised target detection using UAV MPCs is limited. The main challenge lies in utilizing local spectral–spatial information in UAV

Received 2 March 2025; revised 18 May 2025 and 13 June 2025; accepted 30 June 2025. Date of publication 3 July 2025; date of current version 9 July 2025. This work was supported in part by the National Science Fund for Distinguished Young Scholars under Grant 62025107 and in part by the Open Fund Project of KuiYuan Laboratory under Grant KY202423. *(Corresponding author: Xian Li.)*

The authors are with the School of Electronics and Information Engineering, Harbin Institute of Technology, Harbin 150001, China (e-mail: xianli@hit.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3585524

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:49:57 UTC from IEEE Xplore. Restrictions apply.

MPCs without contaminating global statistical backgrounds, especially for occluded targets [13], [14]. Few unsupervised detection methods have been proposed to address the problem of insufficient model training samples for 3-D object detection in MPCs. SF-UDA^{3D} [15] was proposed to perform unsupervised detection on vehicular data using transfer learning methods. The performance of LiDAR-based models is heavily dependent on the density, spatial resolution, and range of the LiDAR. Three-dimensional object detectors have been shown to generalize poorly across domains due to differences such as object sizes and sensor variations [16]. LiDAR-based detectors face domain shift issues when sample data are limited [17]. In contrast to vehicular LiDAR datasets, unsupervised methods based on ULS point clouds primarily focus on applications such as individual tree segmentation [18], [19], which improved on Reed–Xiaoli (RX) detector [20] for anomaly detection. Inspired by anomaly detectors based on hyperspectral imagery (HSI), this offers valuable insights for unsupervised target detection in MPCs from UAVs. Hyperspectral anomaly detectors aim to identify observations with distinct spectral differences from the surrounding background [21], [22]. The main challenge lies in effectively estimating the background, as anomalies lack prior samples [7] and can contaminate the background [23], [24]. Classic methods like RX detector (RXD) rely on statistical models but ignore spatial information [25]. Many improved RXs address background contamination and Gaussian distribution deviations [26], [27], [28], [29]. However, they remain sensitive to contamination. Representation-based methods avoid statistical assumptions, using background dictionaries for background representation [30], [31], [32], [33], [34]. Detection methods based on deep learning have also achieved promising results [35], [36]; Autoencoder (AE) and their variants, such as Auto-AD [21] and robust graph AE (RGAE) [37], automatically extract features and detect anomalies by reconstruction errors; however, facing with the occluded targets detection, the training and pretraining without samples and prior knowledge pose challenges. We found that the local target detection methods have achieved good results without training [38], [39], [40], [41], providing insights for detecting occluded targets in MPCs; however, effectively utilizing the localized spatial–spectral information of unstructured MPCs and achieving unsupervised occluded detection with local remains one of the key challenges to be addressed.

Spherical models and spherical shell models are widely used for point cloud feature extraction and have achieved good results. However, when applied to localized target detection, they have limitations: existing methods overlook the inherent spatial isolation properties of spherical shell models and instead rely on a statistical analysis of all points within the spherical structure, focusing solely on the rotational invariance of points inside the sphere and their adaptability to unstructured data. The rotational invariance of spherical models is widely utilized for feature extraction. SGPCR [42] uses spherical Gaussian functions to represent 3-D point clouds. PRIN and SPRIN [43] utilize adaptive point density for spherical sampling to extract rotation-invariant features. DenseSphere [44] transforms Cartesian points to spherical coordinates, refines sparse clouds, and integrates graph neural networks for detection. PointNet++ [38] constructs local neighborhoods with spherical models for better local structure recognition. ShellConv [45] resolves point order ambiguity with concentric sphere statistics, supporting traditional convolutions. The aforementioned methods utilize spherical models, but when multiple land covers are present within a sphere, statistical analysis using all points can mix target and background points, contaminating background statistics akin to global background estimation. Thus, we can refine the spherical model to construct an MPC-based local spatial–spectral feature anomaly detection model. This approach not only prevents the contamination of global statistical background by target points but also addresses the challenge of limited training samples due to target occlusion.

The focus of this article is to utilize an unsupervised stereo detection method for occluded targets in MPCs of UAVs, without prior knowledge of targets or sample training. Our approach consists of two components: 1) local background isolation and spatial–spectral feature extraction using a spherical shell model and 2) spatial–spectral collaborative representation for targets. The proposed method begins with an adaptive spherical shell approach to obtain each point's local background point set. Next, using the statistical results of the clustering radius, we calculate the multiscale local spatial features and adaptively select background points to avoid pseudoinverse solving issues caused by a rank-deficient background matrix. Finally, the local spectral characteristics of the center points are expressed according to the background matrix, and the resulting expression errors are designed as the outcomes of unsupervised target detection.

The main contributions of this article are as follows.

1) We propose an unsupervised stereo target detection method for occluded targets with MPCs from UAVs. To our knowledge, this is the first 3-D unsupervised occluded target detection method. The main advantage of the proposed method is its robustness to both occluded and exposed targets in airborne scenes.
2) We introduce a spherical shell representation model for adaptive background selection in unstructured MPCs. Leveraging this concept, we design a multiscale spatial feature extraction method that adaptively prevents rank deficiency in the local background matrix, significantly enhancing the model's generality and accuracy.
3) We design a target-level accuracy evaluation method tailored for UAV MPCs with high resolution. By assessing target units from an application perspective rather than data units, this approach prevents inaccurate assessments caused by partial visibility or low response of target structures, leading to missed detections.

The rest of this article is organized as follows. Section II provides a detailed description of the proposed unsupervised detection method based on the spherical shell model for the MPCs. Section III presents the comparison and analysis of the proposed method with state-of-the-art algorithms. Section IV concludes this article.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:49:57 UTC from IEEE Xplore. Restrictions apply.

II. Spherical Shell Model for Unsupervised Occluded Target Detection

An unsupervised stereo target detection framework using a spherical shell representation model has been proposed to overcome the limited target prior challenges of MPC occluded target detection. Leveraging MPCs' spatial–spectral information, our method detects stereo targets without requiring prior training. As illustrated in Fig. 2, the framework consists of three key modules: 1) an adaptive spherical shell model that isolates local targets through background point selection; 2) multiscale spatial feature extraction within spherical neighborhoods; and 3) a collaborative representation model for stereo target detection. The details of the three components are presented in the following.

A. Spherical Shell Model: Adaptive Background Selection

Local targets detection effectively separates targets from background interference, but is inherently sensitive to noise during the construction of the covariance and autocorrelation matrix. Considering the discrete, nonuniform, and rotationally invariant nature of MPCs, conventional local anomaly detectors in raster images, such as HSI, fail to represent the unstructured distribution of MPCs adequately. We propose a fixed-radius spherical shell model to separate the local background adaptively for MPCs, which is designed for the first time for MPC occluded target detection from UAVs. This approach leverages the spatial separation property to avoid global background pollution by the target and effectively characterizes the spatial distances between points. Considering the continuity of shadow boundaries, this approach ensures the selection of local background points with consistent lighting conditions compared to the point under detection, reducing false alarms and missed detections caused by spectral variability from canopy shadows.

The construction of the spherical shell model is illustrated in Fig. 3. For each point p_i of the MPC dataset $\mathbf{P}$, we take the detection point p_i as the sphere center and construct a spherical shell model with radius R_1 and R_2, and the local background point set $\mathbf{P}_{R1} = \{p_j | R_1 > \|p_j - p_i\| > R_2 > 0\}$ in the spherical shell contains k points. The MPCs contain n bands, and we define the normalized spatial characteristics of point p_i as $\text{Spat}_{p_i} = [c_{i1}, c_{i2}, \ldots, c_{im}]$ and reflectance information of p_i as $\text{Spec}_{p_i} = [b_{i1}, b_{i2}, \ldots, b_{in}]$. The spatial–spectral characteristics of p_i can be denoted as $[\text{Spec}_{p_i}, \text{Spat}_{p_i}]$, and the local background matrix $\mathbf{F}_{p_i}$ of p_i, composed of $\mathbf{P}_{R1}$, can be denoted as a matrix consisting of k points with $(m+n)$ features.

Due to the nonuniformity of MPC, a fixed R_1 and R_2 would lead to an indeterminate number of points in $\mathbf{F}_{p_i}$, resulting in different matrix sizes and leading to rank deficiency. An adaptive radius R_1 is defined to ensure a consistent number of elements k in the background matrix $\mathbf{F}_{p_i}$, preventing rank deficiency and the occurrence of pseudoinverse during matrix inversion. The number of eigenvalues $(m+n)$ and the number of points k in $\mathbf{F}_{p_i} = [\text{Spec}_{p_i}, \text{Spat}_{p_i}]$ can be summarized as $k \geq m+n$. Assuming uniform local point cloud density in the MPC, the number of points in the inner sphere and spherical shell is proportional to their volumes. Assuming that there are l points within the inner sphere of radius R_2, the constraints for R_1 can be described as follows:

$$\left(\frac{4}{3}\pi R_1^3 - \frac{4}{3}\pi R_2^3\right) \cdot \frac{l}{\frac{4}{3}\pi R_2^3} = k \geq m+n \tag{1}$$

$$R_1 \geq R_2\left(1 + \frac{m+n}{l}\right)^{\frac{1}{3}}. \tag{2}$$

The local spatial features of p_i are calculated from the coordinates of all points within a certain radius. To ensure maximum isolation between the local background points and the point under detection, we set the radius for spatial feature extraction as the inner sphere radius R_2. The determination of R_2 is elaborated in Section II-B.

B. Local Multiscale Spatial Feature Extraction

Extracting spatial features between targets and backgrounds is conducive to the separation for unsupervised target detection of MPCs. However, single-scale feature extraction struggles with robustness due to varied anomalies and unknown target sizes, as illustrated in Fig. 4(a). This limits the detection of occluded targets. To improve robustness, we introduce a multiscale spatial feature extraction method.

First, we analyzed potential spatial feature differences between 3-D targets and local backgrounds, including distinct normal vectors and concave edges of targets that enhance features like planarity, surface variation, and normal variation. Density contrasts caused by occlusion also improve discrimination. Stereo objects often exhibit broken surfaces, lacking spatial continuity. Guided by these observations, we selected spatial features that are likely more sensitive to targets and listed them in Table I. By performing eigen decomposition, we obtain the corresponding eigenvalues λ_1–λ_3 and eigenvectors, arranged in descending order. Here, $\bar{\mathbf{n}}_z$ is the mean of the vertical components of the normal vectors, l is the number of points in the inner sphere, and $d(p_j, \pi_i)$ is the distance from point p_j to the fit plane π_i and the covariance matrix $\sum$ formed by the center points $\bar{p}$ of $\mathbf{P}_{R2}$

$$\Sigma = \frac{1}{k}\sum_{i=1}^{k}\left(p_i - \bar{p}\right)\left(p_i - \bar{p}\right)^T. \tag{3}$$

Second, to enhance algorithm robustness, we use two different radii to extract multiscale spatial features from MPCs. We set the first radius as $R_3 = \min(r)$, which is the minimum clustering radius r for nonground points, to ensure that the spatial features are representative and that the features of the smallest clusters are not overshadowed. Leveraging the separation characteristics of the spherical shell model from Section II-A, another radius set is R_2, to achieve a more distinct separation between targets and the background. The spatial features for each point p_i are separately calculated for the inner sphere and spherical shell, rather than using the statistical spatial features derived from all points within the sphere. The isolation points of the inner sphere $(p_j | d < R_2)$ preserve the spatial features Spat_{R2} of the central point, and $\text{Spat}_{\text{shell}}$ is used to represent the spatial features of points

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:49:57 UTC from IEEE Xplore. Restrictions apply.

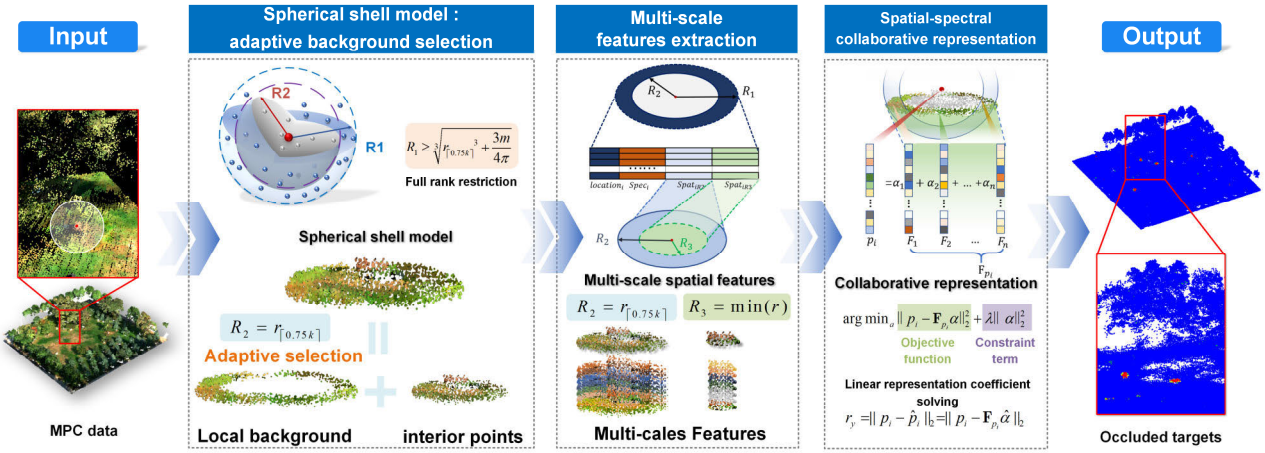
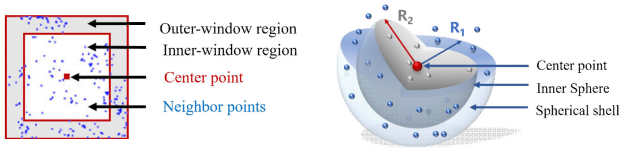
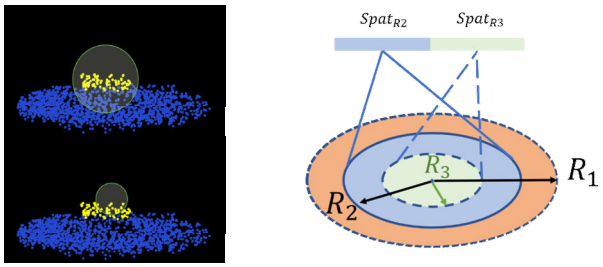


Fig. 2. Overall architecture of the proposed method, including spherical shell model, multiscale features extraction, and spatial–spectral collaborative representation.



(a) (b)

Fig. 3. (a) Dual-window detectors based on imagery cannot be applied to unstructured point cloud data. We utilize the isolation property of (b) spherical shell model for unstructured MPCs to acquire the local background points (blue) of the point under detection (red).



(a) (b)

Fig. 4. Multiscale spatial feature extraction in variable density unstructured MPCs can increase the robustness of the method. (a) Scale affects the feature extraction results. (b) Multiscale feature extraction.

within the spherical shell. Spatial separation avoids contamination of local background by anomalous points, aiding target–background separation in local detection.

Building on the previous chapter, obtaining an adaptive spherical shell model robust to different MPC data is crucial, with determining R_2 being essential. We conducted a comparative verification using three smaller areas of another MPCs containing different targets under the canopy, the spherical shell model based on the RX method with multiscale spatial features at various scales, and evaluated with the area under the curve (AUC). To enhance the method's robustness, the settings of R_2 and R_3 are obtained by statistics of the MPCs. After removing the ground from the point cloud, we cluster the resulting nonground points. For each cluster, we used the radius of the circumscribed sphere obtaining radii $r_i (i = 1, 2, \ldots, m)$ to discuss the influence on the spherical

TABLE I

SUMMARY OF SPATIAL FEATURES

Features	Formula
Planarity (P)	$P(p_i)=\frac{\lambda_2-\lambda_3}{\lambda_1}$
Surface Variation (SV)	$SV(p_i)=\frac{\lambda_3}{\lambda_1+\lambda_2+\lambda_3}$
Omnidirectional (O)	$O(p_i)=(\lambda_1\cdot\lambda_2\cdot\lambda_3)^{\frac{1}{3}}$
Linearity (L)	$L(p_i)=\frac{\lambda_1-\lambda_2}{\lambda_1}$
Verticality (V)	$V(p_i)=\lvert\vec{\mathbf{n}}_z\rvert$
First Moment (M)	$M(p_i)=\left(\frac{1}{l}\sum_{i=1}^{l}x_i,\frac{1}{l}\sum_{i=1}^{l}y_i,\frac{1}{l}\sum_{i=1}^{l}z_i\right)$
Normal Variation (NV)	$NV(p_i)=\frac{\cos^{-1}(\mathbf{n}_i\cdot\vec{\mathbf{n}}_i)}{\lVert p_i-\bar{p}\rVert}$
Information Entropy (IE)	$IE(p_i)=-\sum_{j=1}^{l}p_j\log(p_j)$
Roughness (Ro)	$Ro(p_i)=\frac{1}{l}\sum_{j=1}^{l}\lvert d(p_j,\pi_i)\rvert$

TABLE II

DESCRIPTION OF THE TARGETS IN EACH DATASET

Dataset	Targets	Description	Points	Ratio
HIT Campus 2022 AM	C1, C2, C3, C4,	Exposed cartons	1043	0.133%
	C5	Obscured cartons	139	0.018%
	P1, P2, P3, P4	Stained targets	50	0.006%
HIT Campus 2022 PM	C1,	Exposed cartons	255	0.033%
	C2, C3, C4, C5	Obscured cartons	469	0.061%
	P1, P2, P3, P4	Stained targets	338	0.044%
HIT Campus 2023	T1, T2, T3, T4	Camouflaged tents	7513	6.661%
	T5	Fully obscured tents	1178	1.044%
Houston 2018	A1-A7	Cars	3,347	0.235%

shell model detection with local multiscale spatial features extraction. The results in Fig. 5 show that multiscale spatial features are beneficial for unsupervised target detection, with the highest detection performance achieved when R_2 is close to $r_{\lceil 0.75k \rceil}$ of the radius, and we defined $R_2 = r_{\lceil 0.75k \rceil}$.

The proposed multiscale feature extraction method based on the spherical shell model addresses the uneven point density of occluded targets. The fixed radius leads to varying

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:49:57 UTC from IEEE Xplore. Restrictions apply.

TABLE III
EVALUATION OF INDIVIDUAL TARGETS

Method	HIT 2022 AM	HIT 2022 PM	HIT 2023	Houston 2018
N_C	9	9	5	28
τ_E	0.995	0.998	0.950	0.985
RXD	0.3000	0.0746	0.4032	0.2300
FEBPAD	0.3103	0.0769	0.4166	0.2212
RGAE	0.3103	0.0769	0.4166	0.2277
LSMAD	0.2045	0.0610	0.3246	0.2054
SSRX	0.2195	0.1786	0.3869	0.2460
DCA_CRD	0.6042	0.7811	0.4070	0.5731
DS-RXD	0.1915	0.0382	0.1984	0.2160
proposed	**0.8182**	**0.0909**	**0.5**	**0.8020**

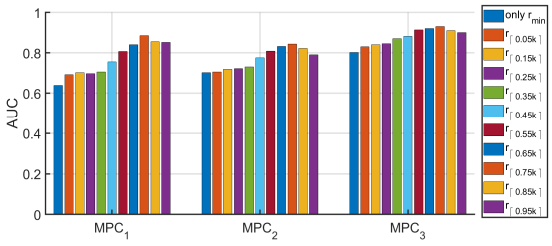


Fig. 5. Effect of multiscale radius on detection accuracy.

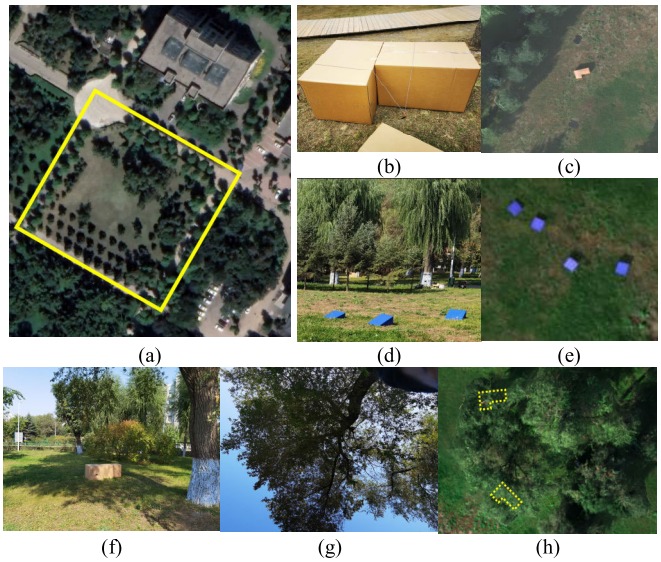


Fig. 6. Overview of HIT Campus 2022 AM/PM datasets. (a) Illustration of the data collection scope in HIT Campus, (b) carton targets, (c) carton targets in an overhead view, (d) stained targets, (e) stained targets in an overhead view, (f) arrangement of canton targets, (g) canopy above the targets, and (h) orthophoto of occluded canton targets where the targets cannot be directly observed, marked with dashed line.

spatial characteristics across different point cloud densities. The spherical shell leverages the rotational invariance of point cloud data, achieving optimal detection when the inner window size is larger than the target size while preventing overshadowing when the size is much larger than the target.

C. Collaborative Representation in MPCs

MPCs sample the land cover continuously, and adjacent points exhibit a certain degree of continuity in their spatial–spectral features. We assume that the spatial–spectral



Fig. 7. Overview of HIT Campus 2023 dataset. (a) Illustration of the data collection scope in HIT Campus. Arrangements of targets: (b) camouflaged tents under the larch forest and (c) tents in an overhead view.



Fig. 8. Overview of 2018 Houston dataset. (a) Illustration of the data collection scope of the Houston dataset and (b) cars in an overhead view.

characteristics of background points can be captured by a linear representation model derived from the background points themselves. In contrast, points of targets cannot be similarly represented. We combined the spectral reflectance Spec_{p_i} recorded at any point p_i in the MPC with its multiscale spatial features $\mathbf{P}_{R2}$.

Consequently, we construct a collaborative representation model to address this issue. The minimization of the ℓ_2-norm enhances the collaborative representation among these points. To adjust the contribution of each neighboring point, a simple yet closed-form solution is proposed, which optimizes using a distance-weighted regularization matrix. By constraining the weight vector such that the sum equals 1, the stability of the optimization effect of the regularization matrix is enhanced with the weights α

$$\arg\min_{a} \left\| p_i - \mathbf{F}_{p_i}\alpha \right\|_2^2 + \gamma \|\alpha\|_2^2. \tag{4}$$

The objective function is simplified with Lagrange multipliers γ as follows:

$$\arg\min_{\alpha} \left[\alpha^{\mathbf{T}}\left(\mathbf{F}_{p_i}^{\mathbf{T}}\mathbf{F}_{p_i} + \gamma\mathbf{I}\right)\alpha - 2\alpha^{\mathbf{T}}\mathbf{F}_{p_i}^{\mathbf{T}} p_i\right] \tag{5}$$

where $\mathbf{I}$ is the identity matrix. However, to further reduce the contamination caused by the target in the background characteristics, we retain $\mathbf{F}_{p_i}$ as $\sum_{i=1}^{k}\lambda_i \geq 0.95\sum_{i=1}^{n}\lambda_i$, where λ_i is the eigenvalue of the covariance matrix of $\mathbf{F}_{p_i}$. This ensures that (6) is minimized and equals zero. Consequently, the expression for α can be derived as follows:

$$\hat{\alpha} = \left(\mathbf{F}_{p_i}^{\mathbf{T}}\mathbf{F}_{p_i} + \gamma\mathbf{I}\right)^{-1}\mathbf{F}_{p_i}^{\mathbf{T}} p_i. \tag{6}$$

Considering the adjustment of the weight vector, a distance-weighted Tikhonov regularization method is adopted. The residual can then be calculated to determine whether the test point is a target

$$D_{\text{DS-SSD}}(p_i) = \|p_i - \hat{p}_i\|_2 = \|p_i - \mathbf{F}_{p_i}\hat{\alpha}\|_2. \tag{7}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:49:57 UTC from IEEE Xplore. Restrictions apply.

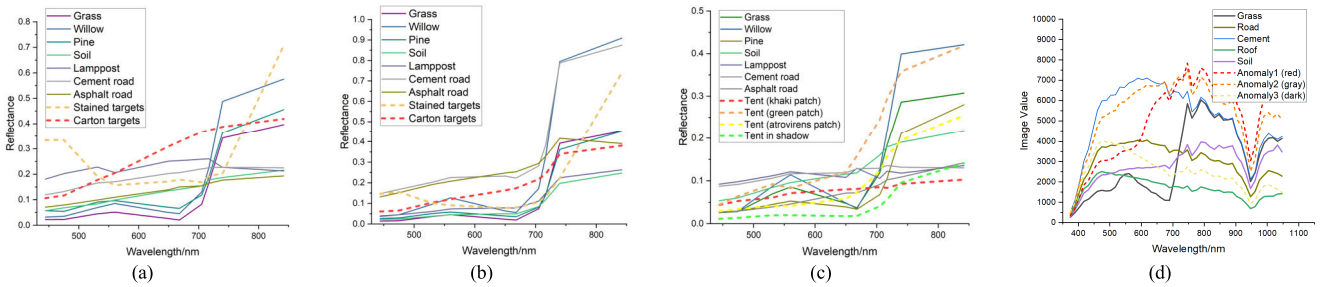


Fig. 9. Spectra of the targets and background of the experimental datasets. (a) 2022 HIT Campus AM dataset. (b) 2022 HIT Campus PM dataset. (c) 2023 HIT Campus dataset. (d) 2018 Houston dataset.

In this method, the local background dictionary requires traversing N points, the number of spatial–spectral features is s, and the overall complexity can be considered as $O(N \cdot \log N) + O(N \cdot s) + O(N \cdot s^3) \approx O(N \cdot s^3)$.

We also use this spherical shell model to enhance the RX method for comparison as dual spherical-based RXD (DS-RXD). It utilizes the covariance matrix $\mathbf{C}$ and mean vector μ of the local background points to construct a second-order matched filter, considering points deviating from the normal distribution as anomalous targets, which can be improved with the spherical shell model as $D_{\text{DS-RXD}}$, and η is used for binary classification. If $D_{\text{DS-RXD}}(p_i) > \eta$, it implies that the result H_0 indicates a background; otherwise, the result H_1 indicates a target

$$D_{\text{DS-RXD}}(p_i) = (p_i - \mu)^T \boldsymbol{C}^{-1}(p_i - \mu) \underset{H_1}{\overset{H_0}{\lessgtr}} \eta. \tag{8}$$

III. Experiments and Analysis

The proposed method was tested on three large-scale UAV-borne fused MPCs and a public airborne dataset. Figs. 6–8 show the scenario of data acquisition and the distribution of targets, Fig. 9 shows the spectral of the targets, and the description of the targets is shown in Table II. It was compared with established detectors, including RXD [20], subspace RX (SSRX) [27], the LRaSMD-based Mahalanobis distance method anomaly detection (LSMAD) [33], RGAE [37], feature extraction and background purification anomaly detector (FEBPAD) [32], the discriminative coefficient analysis-based CRD (DCA_CRD) [34], and DS-RXD. The experimental results are analyzed from both qualitative and quantitative perspectives.

A. Experimental Settings

We validate the method using three large-scale fusion MPCs collected at HIT Campus acquired from the UAV platform. These datasets are all created by fusing LiDAR point clouds and reconstructed point clouds from reflectance multispectral images using Agisoft Metashape, possess a more comprehensive 3-D structure compared to merely surface points generated by image reconstruction, the root mean square error (RMSE) of the registration error being less than 0.5, and have a more complete 3-D structure rather than just surface points generated by image reconstruction, particularly benefiting from the more precise spatial information provided by LiDAR point clouds. Moreover, these datasets demonstrate superior precision in contrast to directly assigning values to point clouds using images.

Algorithm 1 Spherical Shell Collaborative Representation for Unsupervised Stereo Target Detector

Input: For the point cloud dataset P, clustering analysis after removing ground points. Obtain the statistical results of the clustering radii $r_i (i = 1, 2, \ldots, m)$. Gain the R_2, R_3 as well as R_1, and construct the spherical shell model.

for all $P_i \in \mathbf{P}$, **do**

1) $\mathbf{P}_{R1} = \{p_j \mid R_1 > r > R_2 > 0\}$;

2) Calculate the local spatial features with multi-scale $Spat_{p_i} = [Spat_{R_1}, Spat_{R_2}]$ for p_i;

3) Determine the local background corresponding multi-scale spatial-spectral features of p_i; $\mathbf{F}_{p_i} = [Spec_{p_i}, Spat_{p_i}]$

4) Reduce the contamination caused by the target; $\mathbf{F}_{p_i} \leftarrow \sum_{i=1}^{k} \lambda_i \geq 0.95 \sum_{i=1}^{n} \lambda_i, \hat{\alpha} = (\mathbf{F}_{p_i}^{\mathbf{T}} \mathbf{F}_{p_i} + \lambda \mathbf{I})^{-1} \mathbf{F}_{p_i}^{\mathbf{T}} p_i$

5) Distance-weighted Tikhonov regularization method for unsupervised stereo target detection; $D_{DS-SSD}(p_i) = \| p_i - \hat{p}_i \|_2 = \| p_i - \mathbf{F}_{p_i} \hat{\alpha} \|_2$

end

end

1) Dataset Description:

a) HIT campus 2022 AM/PM dataset: These two datasets are a fusion of data obtained from the 3-D reconstruction of ten-band images captured by the Red MX multispectral camera and the point cloud data collected by the Velodyne LiDAR. The data were collected on UAVs on September 27, 2022, at 10:00 A.M. and 4:00 P.M. The targets consist of L-shaped items made of two cartons, securely tied to the grass with a paracord. They measure 0.6 m in height and cover a downward-facing area of 1 m^2. Another type of target consists of blue triangular stained targets positioned horizontally, with dimensions of 0.4 × 0.4 m. The scene includes elm, maple, willow, bald cypress, shrubs, grassland, asphalt, and concrete roads. The distribution of carton targets differs significantly between the two datasets. The HIT Campus 2022 AM dataset contains 781 832 points, with targets located under the trees, while the HIT Campus 2022 PM dataset contains 769 860 points.

b) HIT campus 2023 dataset: This dataset is a fusion of data acquired via UAV, combining the 3-D reconstruction

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:49:57 UTC from IEEE Xplore. Restrictions apply.

from ten-band images taken by the Red MX multispectral camera with the point cloud data gathered by the Reigl-Mini LiDAR. The data were collected on July 7, 2023, when five camouflaged tents were placed under larch forests in HIT Campus. HIT Campus 2023 has 1 015 522 points, including larch, metasequoia, grassland, roads, tents under the forest, and wooden walkways.

c) Airborne houston 2018 dataset: This public dataset was acquired by the Hyperspectral Image Analysis Laboratory and the National Center for Airborne Laser Mapping (NCALM) at the University of Houston. The sensors utilized in this campaign included an Optech Titan MW [46], which collected the point cloud data from a multispectral LIDAR sensor operating at three different laser wavelengths. Additionally, an ITRES CASI 1500 hyperspectral imager was utilized. Hyperspectral data cover a 380–1050-nm spectral range with 48 bands at 1-m ground sample distance. We selected a scene with a variety of features, removed erroneous outliers, combined hyperspectral and point cloud information on the ANN algorithm, and relabeled the targets. The fused data were resampled to 473 385 points using octree, each point with 48 bands. There are a total of 5393 points of targets.

2) Parameters Settings: The RXD and SSRX detectors calculate the entire point cloud with calculate the covariance matrix. For FEBPAD, we set the fractional transform order p, the upper bound of the rank of low-rank part r to 0.7, three for all datasets, and the upper bound of the cardinality of sparse part k 0.7 $\times$ n (where n is the number of points in each dataset). For the RGAE method, we set the tradeoff parameter λ to 0.01, the number of supervoxel S to 150, and the number of hidden layer nodes n_{hid} to 100. The rank r of the LSMAD method is set to 3. For the DCA_CRD method, to be applicable to MPC, the parameters of the W_{inner} and W_{outer} are consistent with the definition of the spherical shell model; we set the positive integer M to 4, the inner window and outer window sizes. We set the outer points parameter as 40 for the proposed method.

In all experiments, the parameters in these compared methods consist of the corresponding reference. The proposed method was coded in Python and run on one core of an Intel Core i5-10400 2.90 GHz CPU, with 32-GB memory, on a Windows 10 system. The detection results of all these datasets are shown in Figs. 10–13.

B. Experimental Results

In this section, the results of the proposed spherical shell collaborative representation method for unsupervised target detection are analyzed and compared with the results obtained by other methods. To visually clarify the unsupervised target detection of occluded targets, we removed points above them in the HIT 2022 AM/PM and HIT 2023 datasets, displaying the scene from an overhead view in Figs. 10–12. The black part is the background, and the exception response value of the target is rendered in red to blue from high to low. Additionally, the 2018 Houston dataset results were shown from a side view in Fig. 13 to highlight targets and set the background in white.

1) HIT Campus 2022 AM Dataset: As the result shown in Fig. 10, all algorithms demonstrated a high detection accuracy. In the campus AM results, we observe that RXD, FEBPAD, and LSMAD are more sensitive to the stained targets P1–P4. Although these methods detect the larger carton targets, their anomaly response values are lower. Conversely, the SSRX algorithm is more sensitive to the carton targets in the scene. Notably, both RGAE and SSRX mark the bright concrete ground in the study area as anomalies. The RGAE results show high response values for shadows, while SSRX is more sensitive to edge points. There is a higher false alarm rate for the edge points in the DCA_CRD result. Comparing the improved DS-RXD algorithm based on the spherical shell model with our proposed algorithm, DS-RXD exhibits more false positives, particularly from the edge points of ground objects and bright objects in the imagery. Our proposed algorithm demonstrates higher target response values, though sparse areas in the point cloud may cause some false alarms.

2) HIT Campus 2022 PM Dataset: As the results shown in Fig. 11, the methods based on global statistical results focus the anomalies on the smaller proportion of targets, such as P1–P4, while the larger targets C1–C5 are overwhelmed by the background, especially in the LSMAD algorithm, where the C1–C5 targets are completely submerged. False alarms mainly come from the cement ground in the background and shadows on the ground in the RGAE and SSRX algorithms. The DCA_CRD result shows that P1–P4 have higher detection accuracy, but there are relatively obvious false alarms around C5 and the road. The anomalies detected by the DS-RXD algorithm are mostly concentrated in some scattered points under the canopy. In contrast, the proposed method identifies anomalous points in the sparser areas under the trees.

3) HIT Campus 2023 Dataset: In this dataset, target T3, being fully exposed to sunlight, demonstrates elevated anomaly response values in RXD, RGAE, and LSMAD. Conversely, FEBPAD and LSMAD show a higher incidence of false positives on the left concrete surface. For obscured tents T4 and T5, responses are minimal. The RGAE's absence of canopy ground spectral data leads to detection resembling estimated textures. DCA_CRD, DS-RXD, and our method show decreased precision for fully occluded or shadowed objects.

4) Houston 2018 Dataset: In this dataset, targets are evenly distributed and both exposed and occluded. However, the integration of HSI with Titan point cloud data suffers from low spatial resolution, causing vertical objects to receive incorrect, identical spectral assignments and spectral mixing. Consequently, occluded cars were assigned tree canopy spectra, resulting in an undetected status in areas A1 and A2 during unsupervised target detection. Despite this, exposed stereo targets had a higher response value. In area A3, dark-colored cars were poorly detected by RXD, FEBPAD, RGAE, and LSMAD, although gray cars were detected. Cars in area A6 were detected by nearly all algorithms. However, dark-colored cars in regions A4, A5, and A7, which spectrally matched the curves of roofs and roads, posed a challenge for global unsupervised detectors caused by a polluted background. Our method, despite being affected by incorrect spectral assignments for occluded cars, enabled them to be

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:49:57 UTC from IEEE Xplore. Restrictions apply.

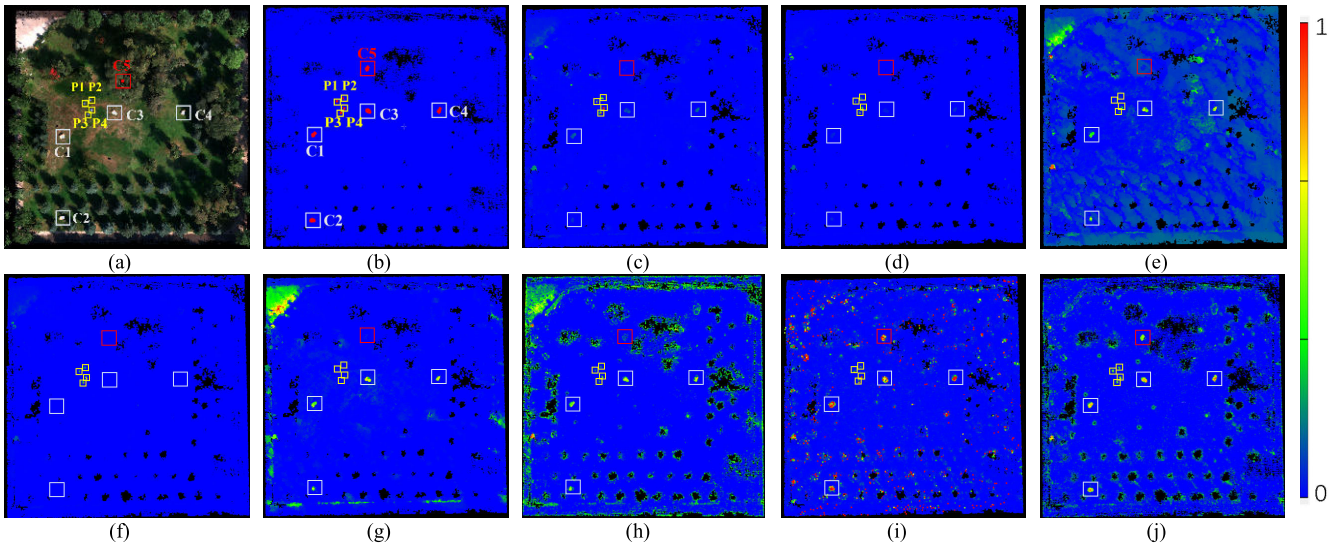


Fig. 10. HIT Campus 2022 AM dataset and the unsupervised detection results. (a) RGB pseudo-color point cloud. (b) Ground-truth map (obstruct removed). (c) RXD. (d) FEBPAD. (e) RGAE. (f) LSMAD. (g) SSRX. (h) DCA_CRD. (i) DS_RXD. (j) Proposed.

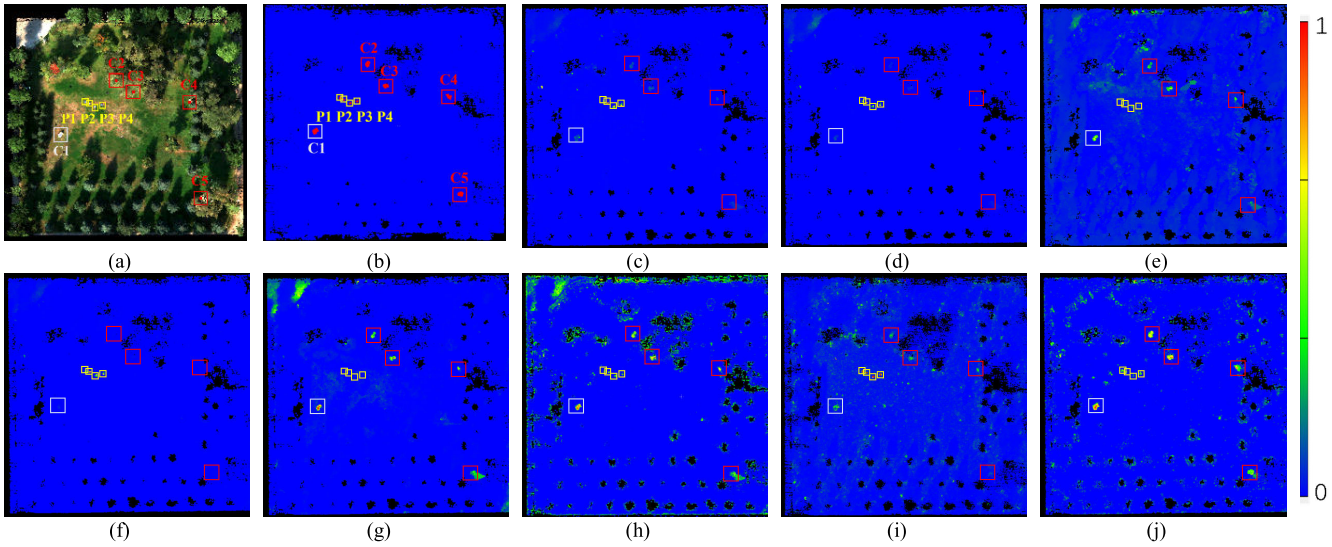


Fig. 11. HIT Campus 2022 PM dataset and the unsupervised detection results. (a) RGB pseudo-color point cloud. (b) Ground-truth map (obstruct removed). (c) RXD. (d) FEBPAD. (e) RGAE. (f) LSMAD. (g) SSRX. (h) DCA_CRD. (i) DS_RXD. (j) Proposed.

distinguished from the background, and a clear difference was observed for exposed targets in area A3.

C. Quantitative Evaluation

In this section, we conduct a quantitative analysis of detection results. An evaluation method evaluation of individual targets (EITs) is proposed for individual target units, which assesses accuracy at the target level. We also utilize standard metrics such as 3-D receiver operating characteristic analysis (3D-ROC) in Figs. 14–17, calculating the AUC, and employing separability maps [21] for unsupervised detection evaluation.

EITs: While metrics like AUC and ROC curves objectively assess performance, practical applications require evaluating detection probability based on individual targets, not single points of MPCs. A target-based evaluation has been proposed for assessing the accuracy of unsupervised stereo detection. We use τ to divide the normalized continuous detection results into abnormal and background and calculate EIT according to the clustering result N_C of abnormal points to evaluate the accuracy of unsupervised stereo detection results. When all abnormal targets are detected, the ratio of the number of anomalies N_G to the total number N_C of detected targets is calculated with values closer to 1 indicating better performance

$$\mathrm{EIT} = \frac{N_G}{\mathrm{ClusterCount}\left(\{\mathbb{I}(R_i > \tau)\}_{i=1}^{|R|}\right)}\Bigg| N_G \subseteq N_C$$
$$\text{s.t. } \tau_{\max} = \arg\max_{\tau}\{N_C(\tau) | N_G \subseteq N_C(\tau)\}. \quad (9)$$

Three Dimensional ROC: Compared to the conventional ROC curves plotted by P_D (detection probability) and P_F (false alarm rate), the 3-D ROC curve introduces the threshold τ with a value range of [0, 1] as a third dimension, which is used for binary segmentation of targets and backgrounds when

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:49:57 UTC from IEEE Xplore. Restrictions apply.

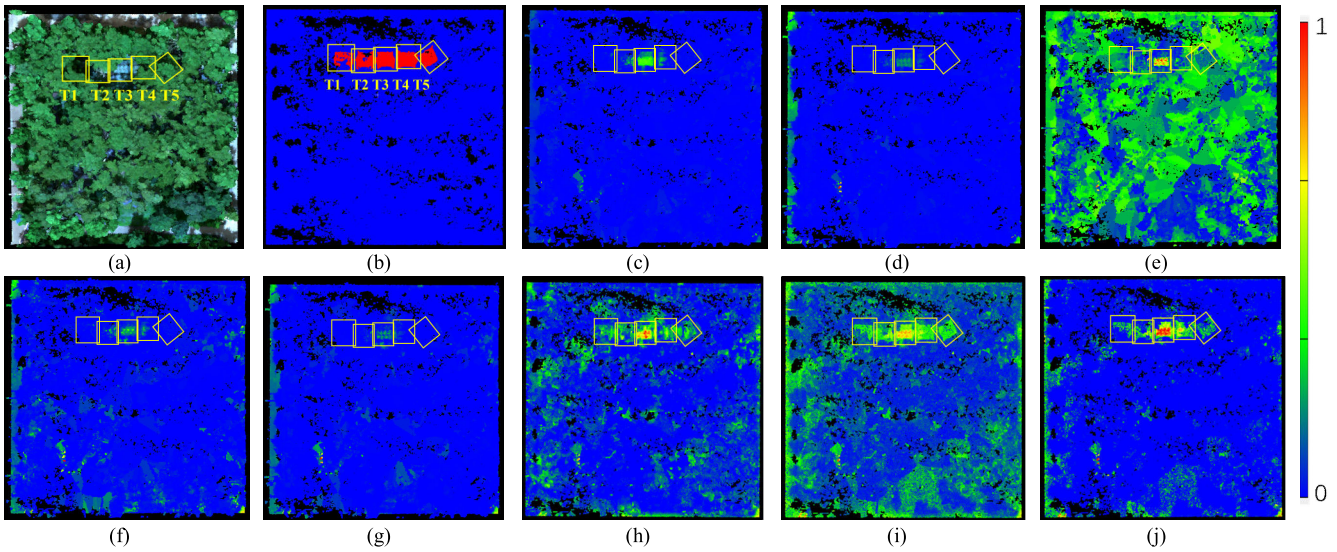


Fig. 12. HIT Campus 2023 dataset and the unsupervised detection results. (a) RGB pseudo-color point cloud. (b) Ground-truth map (obstruct removed). (c) RXD. (d) FEBPAD. (e) RGAE. (f) LSMAD. (g) SSRX. (h) DCA_CRD. (i) DS_RXD. (j) Proposed.

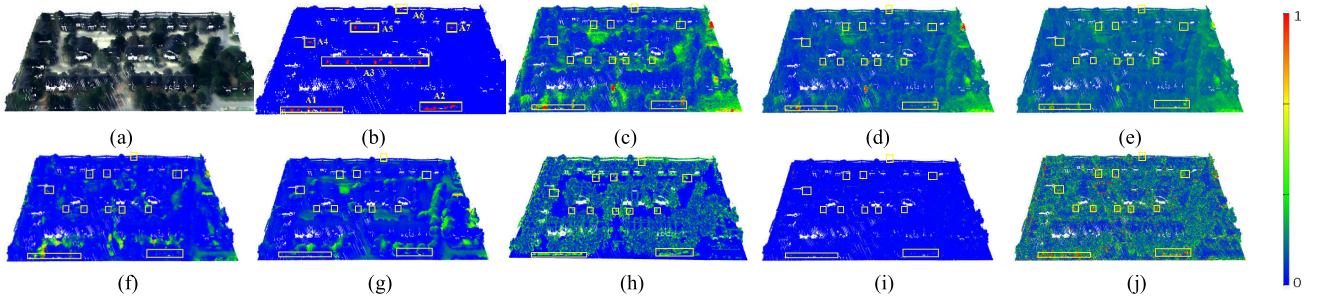


Fig. 13. Houston 2018 dataset and the unsupervised detection results. (a) RGB pseudo-color point cloud. (b) Ground-truth map. (c) RXD. (d) FEBPAD. (e) RGAE. (f) LSMAD. (g) SSRX. (h) DCA_CRD. (i) DS_RXD. (j) Proposed.

calculating consecutive values of P_D and P_F, and enables a more objective evaluation between (P_D, τ) and (P_F, τ) changes

$$P_F = \frac{N_F}{N_{\text{all}}}, \quad P_D = \frac{N_D}{N_{\text{all_truth}}} \tag{10}$$

where N_F is the count of background points incorrectly classified as target points, N_{all} signifies the total number of pixels, N_D denotes the number of correctly detected points, and $N_{\text{all_truth}}$ represents the total target points.

AUC: The AUC of the conventional 2-D ROC curve is employed to evaluate detection outcomes quantitatively. It is calculated as the area enclosed by the curve (P_D, P_F) and the x-axis, with values ranging from 0 to 1. The AUC value serves as a more appropriate evaluation metric. A higher AUC indicates greater detection accuracy, with bold values suggesting the best results and underlined ones showing the second-best outcomes.

Separability Map: Target–background separability maps visually assess detection performance. We use the ground-truth labels to categorize detection results into binary classes. Box plots display response values for targets and backgrounds, as shown in Fig. 18. The red box shows target values within the range [10%, 90%], and the green box shows background values. Whiskers indicate extremes, and the line within each box shows the median value of the point distribution. Effective background separation is indicated when the background box is short and distinct from the target box along the vertical axis, while less overlap indicates better efficacy.

1) HIT Campus 2022 AM Dataset: For the Campus2022AM dataset, as shown in Fig. 14(a), the 3-D ROC curve of the proposed method is above the competing methods. Fig. 15 shows the 2-D ROC curves of (P_D, P_F), and the proposed method achieves the highest score of 0.9998. Similarly, the improvement algorithm DS-RXD based on a double model has also been significantly improved, achieving the second score of 0.9939. The 2-D ROC curves of (P_D, τ) in Fig. 16(a) demonstrate that our algorithm maintains a higher detection probability as the threshold τ, with the area enclosed by the coordinate axes being the largest, indicating that our method has a higher detection probability than other algorithms. In Fig. 17(a), our method shows a rapid decrease in the false alarm rate PF as the threshold τ increases. Curves closer to the origin indicate more severe background compression, corresponding to smaller volumes of the background box in Fig. 18(a). Together, these results suggest that our algorithm can achieve a higher detection probability than other algorithms at a lower false alarm rate and maintain a high detection probability as t increases, indicating stronger robustness. In Fig. 18(a), the distinct target boxes and green background boxes are differentiated. Although RGAE and SSRX show relatively high detection probabilities in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:49:57 UTC from IEEE Xplore. Restrictions apply.

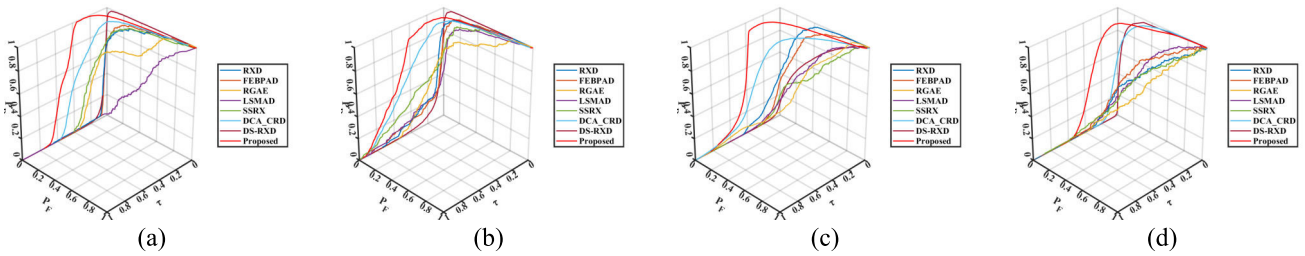


Fig. 14. Three-dimensional ROC curves. (a) HIT Campus 2022 AM dataset. (b) HIT Campus 2022 PM dataset. (c) HIT Campus 2023 dataset. (d) Houston dataset.

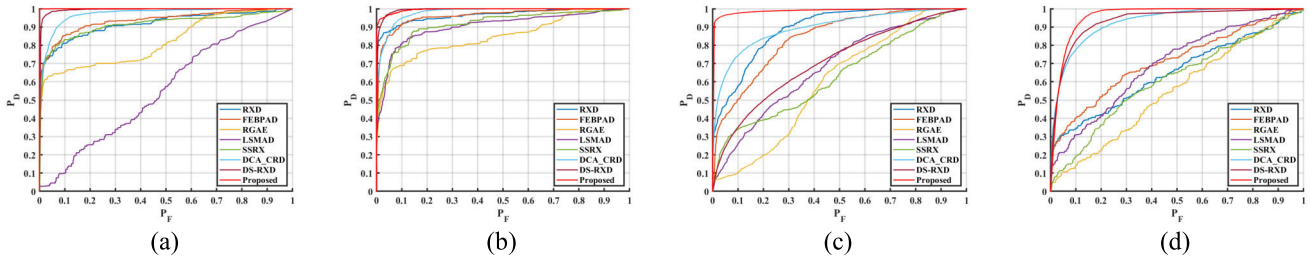


Fig. 15. ROC curves (P_D, P_F). (a) HIT Campus 2022 AM dataset. (b) HIT Campus 2022 PM dataset. (c) HIT Campus 2023 dataset. (d) Houston dataset.

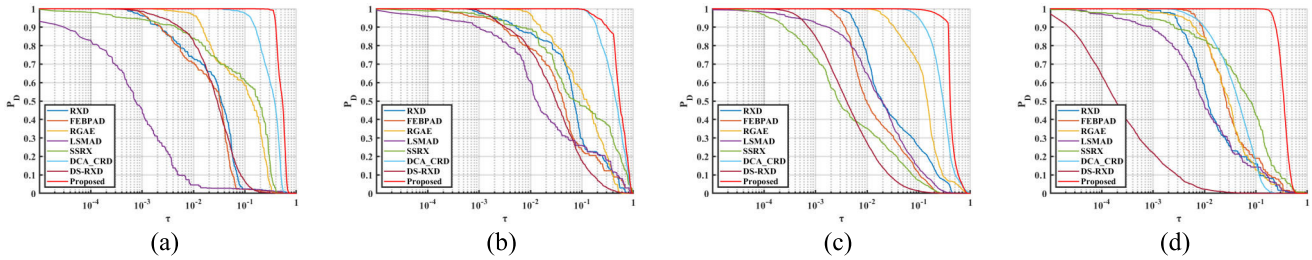


Fig. 16. ROC curves (P_D, τ). (a) HIT Campus 2022 AM dataset. (b) HIT Campus 2022 PM dataset. (c) HIT Campus 2023 dataset. (d) Houston 2018 dataset.

TABLE IV
AUC SCORES OF THE ROC CURVE AND EXECUTION TIME

Method	HIT Campus 2022 AM	HIT Campus 2022 PM	HIT Campus 2023	Houston 2018
RXD	0.9172 (2.19s)	0.9648 (5.27s)	0.8908 (9.70s)	0.6520 (5.61s)
FEBPAD	0.9354 (26.27s)	0.9606 (35.14s)	0.8398 (114.84s)	0.7122 (66.74s)
RGAE	0.8207 (42016.21s)	0.8460 (48999.42s)	0.5988 (84165.54s)	0.5508 (49548.21s)
LSMAD	0.5557 (1.33s)	0.9038 (2.69s)	0.6781 (4.49s)	0.6951 (3.40s)
SSRX	0.9130 (6.70s)	0.9166 (8.94s)	0.6325 (21.16s)	0.6092 (18.85s)
DCA_CRD	0.9718 (4595.21s)	0.9791 (5051.31s)	<u>0.8953</u> (8318.71s)	0.9267 (6027.91s)
DS-RXD	<u>0.9939</u> (2645.10s)	<u>0.9931</u> (2971.02s)	0.7803 (5214.32s)	<u>0.9333</u> (3721.05s)
proposed	**0.9998** (2415.54s)	**0.9953** (3014.42s)	**0.9875** (5024.14s)	**0.9570** (3891.20s)

Fig. 16(a), they perform poorly in Fig. 18(a) due to the influence of P_F, with poor separation between targets and backgrounds. As shown in Table III, our method, which is evaluated by the number of individual targets, performs better in evaluating individual targets at 0.8182.

2) HIT Campus 2022 PM Dataset: For the Campus 2022 PM dataset, Fig. 14(b) shows that our method has a better performance than others. (P_D, P_F) curves in Fig. 15(b) show that our method is closer to the top-left corner of the graph, which shows that our method has the highest AUC value, corresponding to that in Table IV, up to 0.9953. The second method is DB-RXD values in 0.9931. As shown in Figs. 16(b) and 17(b), our method has a higher P_D at a lower P_F, better than SSRX and RGAE. Consistent with the trend of results in Fig. 18(b), our method better separates the background and targets.

3) HIT Campus 2023 Dataset: For the HIT Campus 2023 dataset, the target exhibits not only spatial discontinuity but also spectral variability caused by shadows due to the dense canopy of larch trees obscuring it. As a result, the detection probability is generally poor across various algorithms. As shown in Figs. 16(c) and 17(c), our method has higher at a lower P_F, better than DCA_CRD, SSRX, and RGAE. However, our method demonstrates a relatively higher detection probability.

4) Houston 2018 Dataset: For the Houston dataset, Fig. 15(d) demonstrates that our method outperforms the others, achieving the highest AUC value of 0.9570. The second method only is DS-RXD achieved 0.9333. The incorrect spectral information of occluded car targets as the canopy, increasing the difficulty of unsupervised target detection; this is also the reason for the bad result of (P_D, τ) curves in Fig. 16(d). The curves in Fig. 16(d) indicate that our method yields the largest area under the PD curve relative to the horizontal axis. Additionally, as shown in Fig. 17, our proposed method exhibits rapid convergence of the PF.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:49:57 UTC from IEEE Xplore. Restrictions apply.

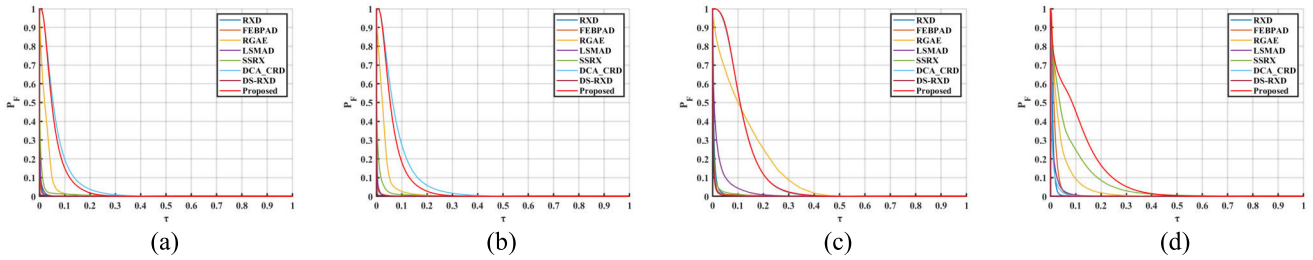


Fig. 17. ROC curves (P_F, τ). (a) HIT Campus 2022 AM dataset. (b) HIT Campus 2022 PM dataset. (c) HIT Campus 2023 dataset. (d) Houston 2018 dataset.

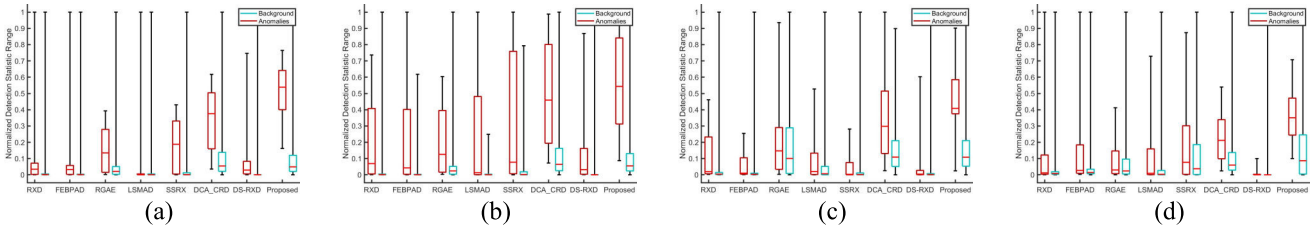


Fig. 18. Separability map. (a) HIT Campus 2022 AM dataset. (b) HIT Campus 2022 PM dataset. (c) HIT Campus 2023 dataset. (d) Houston 2018 dataset.

TABLE V

EVALUATION OF ABLATION EXPERIMENTS FOR UNSUPERVISED TARGET DETECTION BASED ON THE SPHERICAL SHELL MODEL

Module	Background selection		Features selection			AUC
Model	Spherical	Spherical shell	Single scale	Multi scale	Spectral	
1	√			√	√	0.8421
2		√		√		0.7215
3		√			√	0.9466
4		√	√		√	0.9719
5		√		√	√	**0.9953**

Overall, as depicted in Fig. 18, our method more effectively separates the target from the background. In contrast, although the RX and FEBPAD algorithms achieve lower background compression, the lower bounds of the boxes representing the 10% statistical results of the targets do not adequately distinguish from the background median, leading to poorer performance for these two algorithms.

Our proposed method has the highest detection probability in Fig. 16(d). At the same time, in Fig. 17(d), P_F converges rapidly, second only to the DS-RXD algorithm. However, by comprehensive comparison with Fig. 18(d), the target and background of our proposed method can be well separated, so it has a high detection probability, showing our method performs better than others. (P_D, P_F) curves in Fig. 15(d) show that our method has the highest AUC value as 0.9267; the spectral information of the occluded vehicle target is incorrectly assigned as canopy, which increases the difficulty of unsupervised stereo target detection. In Fig. 16(d), our approach achieves the most significant AUC of (P_F, τ) among the compared methods, indicating a broader detection range. Consistent with Fig. 18(d), the proposed method shows superior separation between target and background, as evidenced by the distinct median curves of the target and background. Table IV indicates that our method takes slightly longer to run than global anomaly detection algorithms due to its higher complexity. However, this extra time is meaningful given the substantial improvement in accuracy.

To evaluate the effectiveness of our method, we conducted ablation studies focusing on the spherical shell model, multiscale spatial features extraction, and spatial–spectral features collaborative representation. The experiments contain a complete model and four ablation models, using the HIT Campus 2022 PM dataset. The results are shown in Table V. The results indicate that our proposed method significantly improves the accuracy of detecting occluded targets.

IV. CONCLUSION

We introduce the first unsupervised stereo detection method for occluded targets in UAV MPCs, addressing challenges like limited samples, training difficulties, and occlusion-induced information loss. Our method, which avoids sample training and leverages the spherical shell model's local separation characteristics, adaptively selects local backgrounds and extracts multiscale features to enhance robustness. It outperforms other unsupervised multispectral detectors by accurately detecting both occluded and exposed targets, offering an efficient solution for target detection without sample training. Future work will focus on optimizing the method's complexity to improve efficiency and tackle shadowed area detection in MPCs.

REFERENCES

[1] O. Elharrouss, K. Hassine, A. Zayyan, Z. Chatri, S. Al-Maadeed, and K. Abualsaud, "3D point cloud for objects and scenes classification, recognition, segmentation, and reconstruction: A review," *Cloud Comput. Data Sci.*, vol. 4, no. 2, pp. 134–160, Jun. 2023, doi: 10.37256/ccds.4220232722.

[2] Q. Wang et al., "Unsupervised domain adaptation for cross-scene multispectral point cloud classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5705115, doi: 10.1109/TGRS.2024.3423759.

[3] Y. Hu, J. Xiao, L. Liu, L. Zhang, and Y. Wang, "Detection of small impact craters via semantic segmenting lunar point clouds using deep learning network," *Remote Sens.*, vol. 13, no. 9, p. 1826, May 2021, doi: 10.3390/rs13091826.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:49:57 UTC from IEEE Xplore. Restrictions apply.

[4] S. R. Levick, T. Whiteside, D. A. Loewensteiner, M. Rudge, and R. Bartolo, "Leveraging TLS as a calibration and validation tool for MLS and ULS mapping of savanna structure and biomass at landscape-scales," *Remote Sens.*, vol. 13, no. 2, p. 257, Jan. 2021, doi: 10.3390/rs13020257.

[5] C. W. Ross, E. L. Loudermilk, N. Skowronski, S. Pokswinski, J. K. Hiers, and J. O'Brien, "LiDAR voxel-size optimization for canopy gap estimation," *Remote Sens.*, vol. 14, no. 5, p. 1054, Feb. 2022, doi: 10.3390/rs14051054.

[6] C. Grönwall, T. Chevalier, G. Tolt, and P. Andersson, "An approach to target detection in forested scenes," *Proc. SPIE*, vol. 6950, Apr. 2008, Art. no. 69500S, doi: 10.1117/12.777042.

[7] S. Shokirov et al., "Multi-platform LiDAR approach for detecting coarse woody debris in a landscape with varied ground cover," *Int. J. Remote Sens.*, vol. 42, no. 24, pp. 9324–9350, Dec. 2021, doi: 10.1080/01431161.2021.1995072.

[8] B. Vandendaele, R. A. Fournier, U. Vepakomma, G. Pelletier, P. Lejeune, and O. Martin-Ducup, "Estimation of northern hardwood forest inventory attributes using UAV laser scanning (ULS): Transferability of laser scanning methods and comparison of automated approaches at the tree- and stand-level," *Remote Sens.*, vol. 13, no. 14, p. 2796, Jul. 2021, doi: 10.3390/rs13142796.

[9] G. Mandlburger et al., "Complementing airborne laser bathymetry with UAV-based LiDAR for capturing alluvial landscapes," *Proc. SPIE*, vol. 9637, pp. 60–73, Oct. 2015, doi: 10.1117/12.2194779.

[10] W. Zhou et al., "UAV laser scanning technology: A potential cost-effective tool for micro-topography detection over wooded areas for archaeological prospection," *Int. J. Digit. Earth*, vol. 13, no. 11, pp. 1279–1301, Nov. 2020, doi: 10.1080/17538947.2019.1711209.

[11] Z. Xu et al., "Tree species classification using UAS-based digital aerial photogrammetry point clouds and multispectral imageries in subtropical natural forests," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 92, Oct. 2020, Art. no. 102173, doi: 10.1016/j.jag.2020.102173.

[12] Z. Qin, X. Li, and Y. Gu, "Hemisphere harmonics basis: A universal approach to remote sensing BRDF approximation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5403412, doi: 10.1109/TGRS.2024.3374237.

[13] T. Arevalo-Ramirez and F. A. Cheein, "Predicting the elevation of canopy occluded ground points in dense forest regions," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5619110, doi: 10.1109/TGRS.2022.3152925.

[14] X. Li, M. Ding, and A. Pižurica, "Deep feature fusion via two-stream convolutional neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2615–2629, Apr. 2020, doi: 10.1109/TGRS.2019.2952758.

[15] C. Saltori, S. Lathuilière, N. Sebe, E. Ricci, and F. Galasso, "SF-UDA3D: Source-free unsupervised domain adaptation for LiDAR-based 3D object detection," in *Proc. Int. Conf. 3D Vis. (3DV)*, Nov. 2020, pp. 771–780, doi: 10.1109/3DV50981.2020.00087.

[16] M. Hansson, "Unsupervised domain adaptation for 3D object detection using adversarial adaptation: Learning transferable LiDAR features for a delivery robot," Ph.D. dissertation, KTH Royal Inst. Technol., Stockholm, Sweden, 2023.

[17] Y. Wang et al., "Train in Germany, test in the USA: Making 3D object detectors generalize," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Germany, WA, USA: Making 3D Object Detectors Generalize, Jun. 2020, pp. 11710–11720, doi: 10.1109/CVPR42600.2020.01173.

[18] X. Chen, K. Jiang, Y. Zhu, X. Wang, and T. Yun, "Individual tree crown segmentation directly from UAV-borne LiDAR data using the PointNet of deep learning," *Forests*, vol. 12, no. 2, p. 131, Jan. 2021, doi: 10.3390/f12020131.

[19] Z. Liang, Z. Li, S. Xu, M. Tan, and K. Jia, "Instance segmentation in 3D scenes using semantic superpoint tree networks," presented at the IEEE/CVF Int. Conf. Comput. Vis. (ICCV), Oct. 2021.

[20] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Aug. 1990, doi: 10.1109/29.60107.

[21] S. Wang, X. Wang, L. Zhang, and Y. Zhong, "Auto-AD: Autonomous hyperspectral anomaly detection network based on fully convolutional autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503314, doi: 10.1109/TGRS.2021.3057721.

[22] X. Zhao, K. Liu, K. Gao, and W. Li, "Hyperspectral time-series target detection based on spectral perception and spatial–temporal tensor decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5520812, doi: 10.1109/TGRS.2023.3307071.

[23] Z. Wu, W. Zhu, J. Chanussot, Y. Xu, and S. Osher, "Hyperspectral anomaly detection via global and local joint modeling of background," *IEEE Trans. Signal Process.*, vol. 67, no. 14, pp. 3858–3869, Jul. 2019, doi: 10.1109/TSP.2019.2922157.

[24] S. Liu, L. Zhang, Y. Cen, L. Chen, and Y. Wang, "A fast hyperspectral anomaly detection algorithm based on greedy bilateral smoothing and extended multi-attribute profile," *Remote Sens.*, vol. 13, no. 19, p. 3954, Oct. 2021, doi: 10.3390/rs13193954.

[25] J. G. R. Ayuga, R. M. Marín, M. M. Sacristán, J. Bonatti, and J. C. Ojeda, "Hyperspectral anomaly detection in urban scenarios," *ISPRS Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. XLI-B7, pp. 111–116, Jun. 2016, doi: 10.5194/isprsarchives-xli-b7-111-2016.

[26] L. Ren, L. Zhao, and X. Li, "Anomaly detection-oriented band selection for hyperspectral image," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 3788–3791, doi: 10.1109/IGARSS.2019.8898990.

[27] A. Schaum, "Joint subspace detection of hyperspectral targets," in *Proc. IEEE Aerosp. Conf.*, Mar. 2004, p. 1824, doi: 10.1109/AERO.2004.1367963.

[28] J. Zhou, C. Kwan, B. Ayhan, and M. T. Eismann, "A novel cluster kernel RX algorithm for anomaly and change detection using hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 11, pp. 6497–6504, Nov. 2016, doi: 10.1109/TGRS.2016.2585495.

[29] N. Billor, A. S. Hadi, and P. F. Velleman, "BACON: Blocked adaptive computationally efficient outlier nominators," *Comput. Statist. Data Anal.*, vol. 34, no. 3, pp. 279–298, Sep. 2000.

[30] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015, doi: 10.1109/TGRS.2014.2343955.

[31] L. Xiong, X. Chen, and J. Schneider, "Direct robust matrix factorizatoin for anomaly detection," in *Proc. IEEE 11th Int. Conf. Data Mining*, Dec. 2011, pp. 844–853, doi: 10.1109/ICDM.2011.52.

[32] Y. Ma, G. Fan, Q. Jin, J. Huang, X. Mei, and J. Ma, "Hyperspectral anomaly detection via integration of feature extraction and background purification," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 8, pp. 1436–1440, Aug. 2021, doi: 10.1109/LGRS.2020.2998809.

[33] Y. Zhang, B. Du, L. Zhang, and S. Wang, "A low-rank and sparse matrix decomposition-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1376–1389, Mar. 2016, doi: 10.1109/TGRS.2015.2479299.

[34] Y. Yang, Q. Yang, S. Song, D. Liu, and J. Zhang, "Discriminative coefficient analysis-based collaborative representation with enhanced background-anomaly separation for hyperspectral anomaly detection," *Infr. Phys. Technol.*, vol. 141, Sep. 2024, Art. no. 105476, doi: 10.1016/j.infrared.2024.105476.

[35] X. Hu et al., "Hyperspectral anomaly detection using deep learning: A review," *Remote Sens.*, vol. 14, no. 9, p. 1973, Apr. 2022, doi: 10.3390/rs14091973.

[36] C. Li, B. Zhang, D. Hong, X. Jia, A. Plaza, and J. Chanussot, "Learning disentangled priors for hyperspectral anomaly detection: A coupling model-driven and data-driven paradigm," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 4, pp. 6883–6896, Apr. 2025, doi: 10.1109/TNNLS.2024.3401589.

[37] G. Fan, Y. Ma, X. Mei, F. Fan, J. Huang, and J. Ma, "Hyperspectral anomaly detection with robust graph autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511314, doi: 10.1109/TGRS.2021.3097097.

[38] Q. Guo, Y. Cen, L. Zhang, Y. Zhang, and Y. Huang, "Hyperspectral anomaly detection based on spatial–spectral cross-guided mask autoencoder," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 9876–9889, 2024, doi: 10.1109/JSTARS.2024.3393995.

[39] Q. Guo, Y. Cen, L. Zhang, Y. Zhang, S. Hu, and X. Liu, "Robust multi-stage progressive autoencoder for hyperspectral anomaly detection," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 134, Nov. 2024, Art. no. 104200, doi: 10.1016/j.jag.2024.104200.

[40] X. Zhao et al., "Tensor adaptive reconstruction cascaded with global and local feature fusion for hyperspectral target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 607–620, 2025, doi: 10.1109/JSTARS.2024.3486774.

[41] X. Zhao, J. Huang, Y. Gao, and Q. Wang, "Hyperspectral target detection based on prior spectral perception and local graph fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 13936–13948, 2024, doi: 10.1109/JSTARS.2024.3439560.

[42] D. Salihu and E. Steinbach, "SGPCR: Spherical Gaussian point cloud representation and its application to object registration and retrieval," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2023, pp. 572–581, doi: 10.1109/WACV56688.2023.00064.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:49:57 UTC from IEEE Xplore. Restrictions apply.

[43] Y. You et al., "PRIN/SPRIN: On extracting point-wise rotation invariant features," 2021, *arXiv:2102.12093*.
[44] J. W. Jung, J. H. Yoon, and S. B. Yoo, "DenseSphere: Multimodal 3D object detection under a sparse point cloud based on spherical coordinate," *Expert Syst. Appl.*, vol. 251, Oct. 2024, Art. no. 124053, doi: 10.1016/j.eswa.2024.124053.
[45] Z. Zhang, B. Hua, and S. Yeung, "ShellNet: Efficient point cloud convolutional neural networks using concentric shells statistics," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1607–1616, doi: 10.1109/ICCV.2019.00169.
[46] L. Chen, Y. Gu, X. Li, X. Zhang, and B. Liu, "A normalized spatial–spectral supervoxel segmentation method for multispectral point cloud data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5704311, doi: 10.1109/TGRS.2023.3313734.



Likun Chen (Member, IEEE) received the M.S. degree in surveying and mapping engineering from Chengdu University of Technology, Chengdu, China, in 2019. He is currently pursuing the Ph.D. degree in information and communication engineering with Harbin Institute of Technology (HIT), Harbin, China.

His research interests include hyperspectral images and LiDAR data processing, and their application in remote sensing.



Yanfeng Gu (Senior Member, IEEE) received the Ph.D. degree in information and communication engineering from Harbin Institute of Technology (HIT), Harbin, China, in 2005.

He joined the School of Electronics and Information Engineering, HIT, as a Lecturer. He was appointed as an Associate Professor at the same institute in 2006; meanwhile, he was enrolled in the first Outstanding Young Teacher Training Program of HIT. From 2011 to 2012, he was a Visiting Scholar with the Department of Electrical Engineering and Computer Science, University of California at Berkeley, Berkeley, CA, USA. He is currently a Professor with the Department of Information Engineering, HIT. He has published more than 100 peer-reviewed papers, and four book chapters, and he is the inventor or co-inventor of 20 patents. His research interests include space-intelligent remote sensing and information processing, multimodal hyperspectral remote sensing, and spaceborne time-series image processing.



Xian Li (Member, IEEE) received the Ph.D. degree in instrument science and technology from Harbin Institute of Technology (HIT), Harbin, China, in 2021.

From 2018 to 2020, he was a Doctoral Researcher with the Department of Telecommunications and Information Processing, Ghent University, Ghent, Belgium, supported by China Scholarship Council. He is currently an Assistant Professor with the School of Electronics and Information Engineering, HIT. His research interests include deep learning, hyperspectral remote sensing, and image analysis.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 04:49:57 UTC from IEEE Xplore. Restrictions apply.